

DailyKhata

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HCI & COMPUTER GRAPHICS (SPRING 2026)

May 2026 — Total Spent

Rs. 28,400

Budget: Rs. 35,000

Remaining: Rs. 6,600



81% of budget used



SCAN



VOICE



TYPE



STATS



SCAN



VOICE

Recent Receipts

[VIEW ALL](#)

Recent Receipts

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Recent Receipts

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DAILY KHATA:
MILESTONE 03

**Initial Prototype &
User Test Plans**

May 2026 — Total Spent

Rs. 28,400

Budget: Rs. 35,000

Remaining: Rs. 6,600



Utility Store — Gulberg

11:30 AM

Tag: Groceries

11:30 AM

12 items

Yesterday

TEAM PATCHES

Saad Mughal | F2023-009

Khadijah Zahoor | F2023-956

SECTION 00

NEEDS FINDING & INITIAL CONCEPT

The Needs Finding Methodology

Study Goals

- Explore how users track daily expenses
- Pinpoint friction causing abandonment
- Understand mental models for categorizing physical receipts
- Gauge attitudes toward AI/voice input



Diverse Profiles Spanning Analogue Traditions and Digital Frustrations

Group A - Homemakers & Shopkeepers			
ID	Profile	Digital Literacy	Current Method
P1	Homemaker, 40s	Low digital literacy	Physical notebook (Khata)
P2	Homemaker, 30s	Mod digital literacy	Notebook + WhatsApp notes
P3	Head of household, 50s	Low digital literacy	Mental tracking + receipts in drawer
P4	Local shopkeeper, 35	Mod digital literacy	Paper ledger
P5	Homemaker, 28	Mod digital literacy	Combination of notes app + notebook
Group B - Students & Freelancers			
P6	Uni student, 21	High digital literacy	Abandoned app (Wallet); now spreadsheet
P7	Uni student, 22	High digital literacy	No consistent system
P8	Freelancer, 26	High digital literacy	Notion + manual receipt photos
P9	SME owner, 30	Mod digital literacy	WhatsApp messages to self
P10	Uni student, 23	Mod digital literacy	Abandoned Cashew app; now mental tracking

AFFINITY DIAGRAM

Receipt Management Issues

Lost it on the way home

Can't read the number

Don't bother saving it

Expense Categorization Problems

No ration category

Groceries doesn't fit my spending

I don't think like that

Manual Input Frustration

Takes too long

Gave up after a week

Feels like extra homework

Cash Transaction Tracking Gaps

Rickshaw has no receipt

Vendor just says the amount

No way to log this

AI Transparency & Trust

What if it reads wrong

I need to double-check

Show me what it understood

Identifying the Friction: Why Current Practices Fail



Practice

Crumpled in purses, tossed in drawers.

Need

A system that digitizes data before the physical paper is lost.



Practice

Abandoning apps because typing every purchase feels like a 'second job.'

Need

Zero-typing data entry.

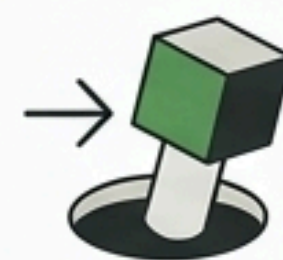


Practice

Universally skipping roadside stalls and rickshaws (30-40% of spending).

Need

A logging method for transactions with zero paper trail.



Practice

Thinking in 'occasions' (weekly bazar) rather than Western app categories.

Need

Flexible, locally-contextualized tagging.



Practice

Refusing to rely on automation if it can't handle mixed Urdu/English.

Need

Absolute transparency and easy error-correction.

Balancing Ambition with Reality: Requirements & Constraints

Functional Requirements (The Ambition)

-  - **Core:** Camera pipeline for handwritten/faded/mixed-language receipts. Voice input for informal cash. Inline verification step (touch-to-edit).
-   - **Secondary:** Flexible local categories. Zero-effort weekly/monthly plain-language summaries.
-   - **Optional/Pro:** Offline capability for low-connectivity areas.



Identified Constraints (The Reality)

- **Technical:** Must run smoothly on low-end Android 9+ devices with 2GB RAM. Background sync for offline logging.  
- **Usability:** Zero mandatory account creation at launch. 'Misc' must always be a valid, unforced bucket.  
- **Financial:** No subscription paywall on core features (highly price-sensitive demographic). 

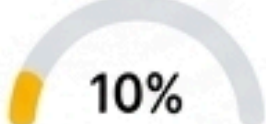
Designing for Two Distinct Realities

Rukhsana Bibi (Ammi Jaan - 43)

 Lahore Homemaker | 40k/mo budget | Android (3GB RAM).

 **Current Tool**
Physical Khata notebook. Receipts in a drawer.

 **Digital Literacy**
 20%
Low - WhatsApp only

 **Tech Trust**
 10%
Distrustful of AI

 **Pain Point**
Faded receipts, forgets small items, apps feel “too English”.

“Main bas ye jaanna chahti hoon ke hafte mein kitna gaya.”

Zaid Malik (The Hustle Student - 22)

 Final-year CS student & Freelancer | Android (4GB RAM).

 **Current Tool**
Notion table (abandoned Cashew app).

 **Digital Literacy**
 95%
High

 **Tech Trust**
 80%
Wants transparency and Pro tools

 **Pain Point**
Categorization takes too long; mixed PKR/foreign gig income.

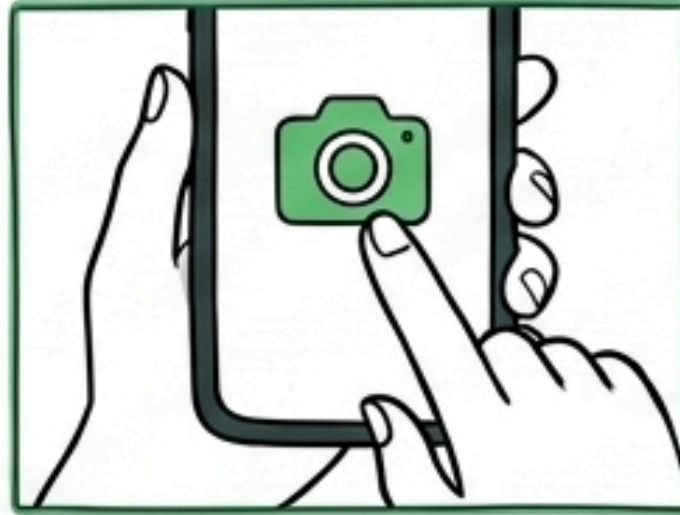
“I want to spend 5 seconds logging, not 5 minutes.”

Storyboards: Frictionless Multimodal Entry

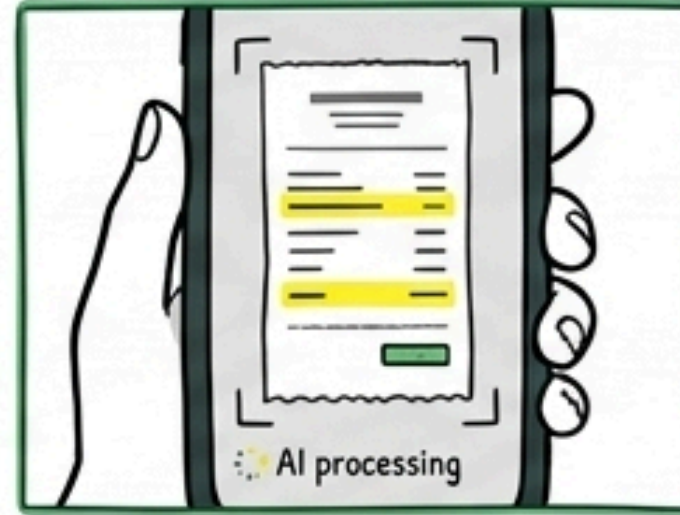
Camera Scan with Verification



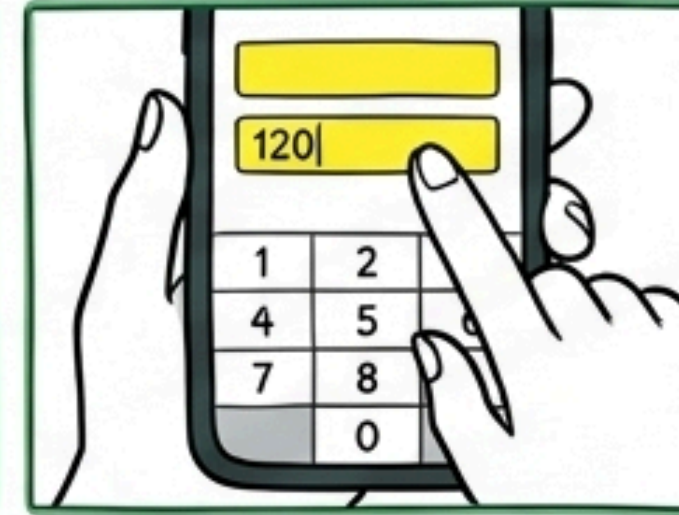
User returns with messy receipt.



One tap on home screen camera icon.



AI processes; unsure fields highlighted in bright yellow.



User taps yellow field, uses numpad to correct.

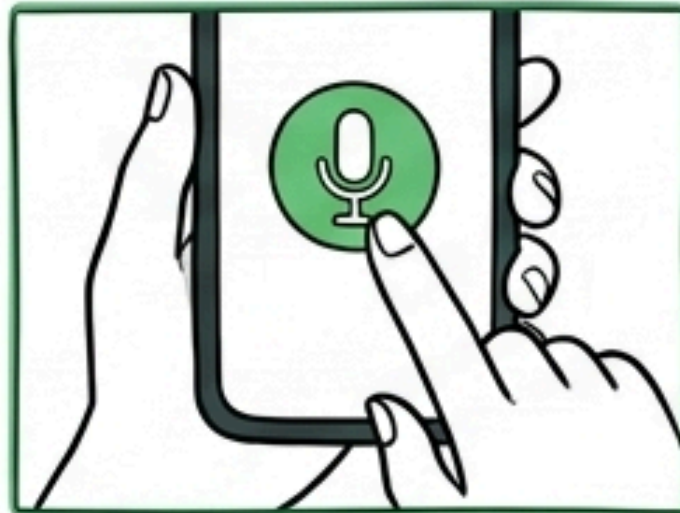


Taps 'Confirm'. Daily total updates. (<30 seconds).

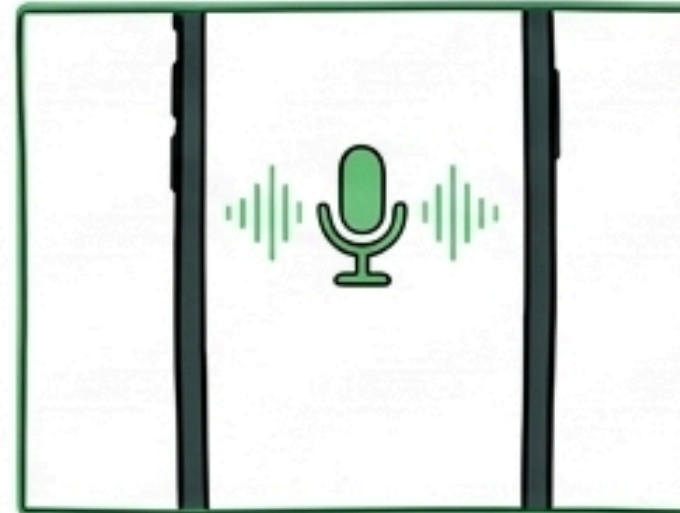
Voice Logging for Informal Cash



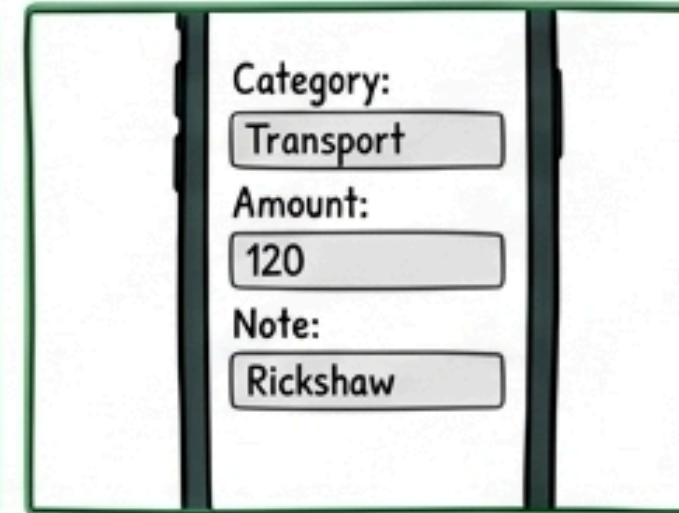
User pays rickshaw driver cash.



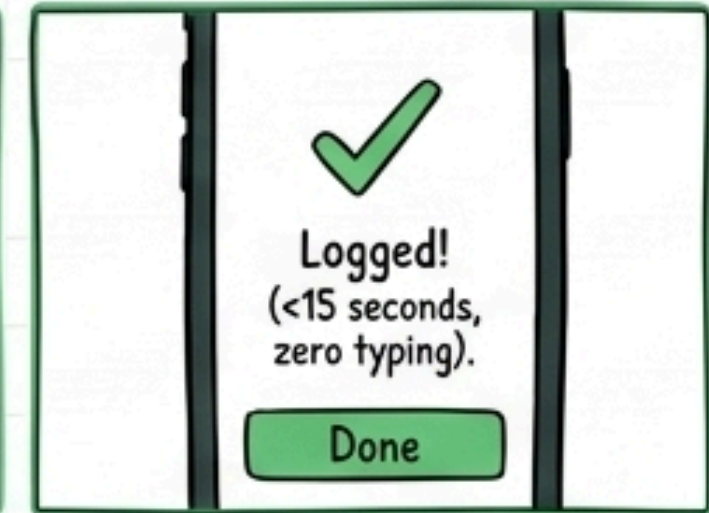
Taps giant microphone icon.



Speaks naturally: "Spent 120 on rickshaw."



Screen shows parsed intent.



Taps 'Confirm'. Done. (<15 seconds, zero typing).

Competitive Matrix: Why Existing Solutions Fail Pakistan

Features	Khata Notebook	Bank App	Cashew	Wallet	Notion	DailyKhata
Receipt Scanning	✗	✗	Basic OCR	Basic OCR	✗	AI (handwritten/mixed)
Voice Input	✗	✗	✗	✗	✗	✓
Informal Cash Logging	Manual	✗	Manual	Manual	Manual	One-tap voice
Urdu/Local Receipt Support	✓	✗	✗	✗	N/A	✓
Inline Verification	N/A	N/A	Auto-commit	Auto-commit	Manual	Edit before save
Local Categories	User-defined	N/A	Western	Western	User-defined	Local defaults
Offline Functionality	✓	✗	Partial	Partial	✓	✓
Onboarding Complexity	None	High	Medium	High	High	Minimal (<60s)

****Only **DailyKhata** supports Voice Input (NLP) Urdu/Local Receipt Support.**
Wallet/Cashew force Auto-commit; **DailyKhata uniquely uses Inline edit before save.**

SECTION 01

PROTOTYPE DESIGN & EXECUTION

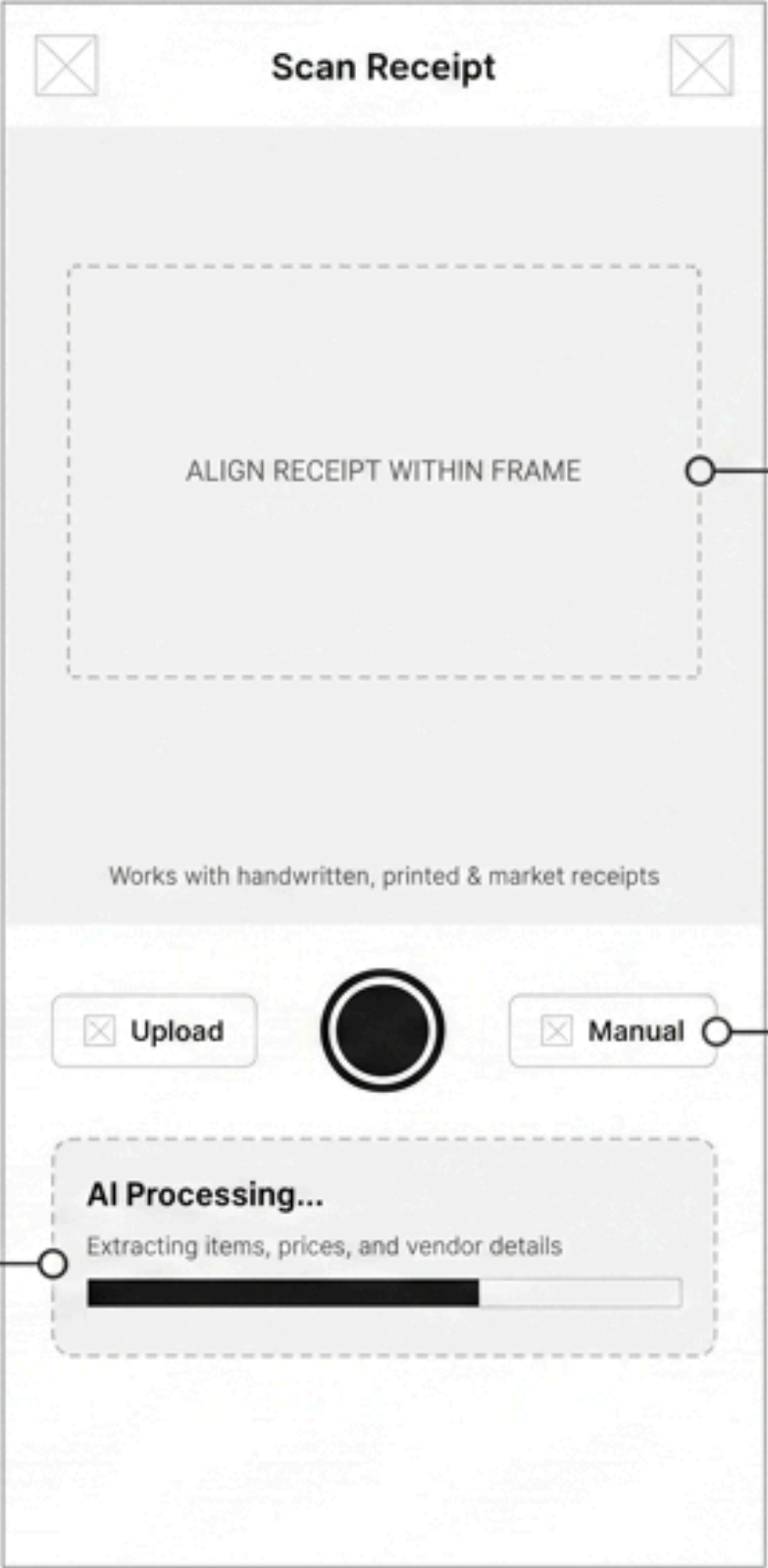
Translating user needs into a low-fidelity, fully interactive, offline-first architecture.

PROTOTYPE SCOPE & FIDELITY

PROTOTYPE GOALS	INCLUSION / EXCLUSION	DATA CONTEXT
● Evaluate AI receipt verification pipeline. ● Assess comprehension of Analytics/Price History. ● Test multimodal entry (Scan/Voice/Manual).	IN SCOPE: ● 6 primary task flows ● Interactive error paths ● Low-fidelity grayscale UI (focus on layout/logic over aesthetics) OUT OF SCOPE: ● Live OCR processing ● Cloud syncing ● Account creation	Hardcoded Pakistani-context dummy data to simulate realistic usage. ● Rs. 28,400 monthly spend ● Utility Store, Atta, LESCO

PROTOTYPE LINK: Active Figma prototype bridging all 13 core screens into a cohesive experience.

UI ANATOMY: MULTIMODAL FRICTIONLESS INPUT



Explicit 'AI Processing' state manages user expectations regarding latency.

Viewfinder designed specifically for handwritten/faded market receipts.

Fallback 'Manual' entry integrated at root level to prevent dead ends.



'Listening' badge ensures system status visibility.

Live transcript builds system trust.

AI-detected item cards populate in real-time, allowing instant verification before saving.

UI ANATOMY: MANDATORY INLINE VERIFICATION

DESIGN RATIONALE

Users fundamentally distrust AI with financial data.

Auto-commit is dangerous; it is replaced with a mandatory review step to ensure data integrity.



Verify Receipt

Save

Utility Store — Gulberg, Lahore

Receipt Date: 16 May 2026 - 11:30 AM

1000 · Food & Groceries

ITEM	QTY	UNIT	TOTAL
Atta (10kg)	1	212	Rs. 212
Doodh (1L)	4	180	Rs. 720
Chawal (5kg)	1	340	Rs. 340
Cooking Oil	2	480	Rs. 960
Total			Rs. 2,232

3 fields flagged — tap to correct before saving

Price Changes Since Last Visit

Atta: Rs. 180 → Rs. 212 +18%

Doodh: Rs. 175 → Rs. 180 +3%

CONFIRM & SAVE TO KHATA

EDIT ITEMS

UNCERTAINTY FLAGS

Fields with low AI confidence are explicitly highlighted with a dashed yellow border.

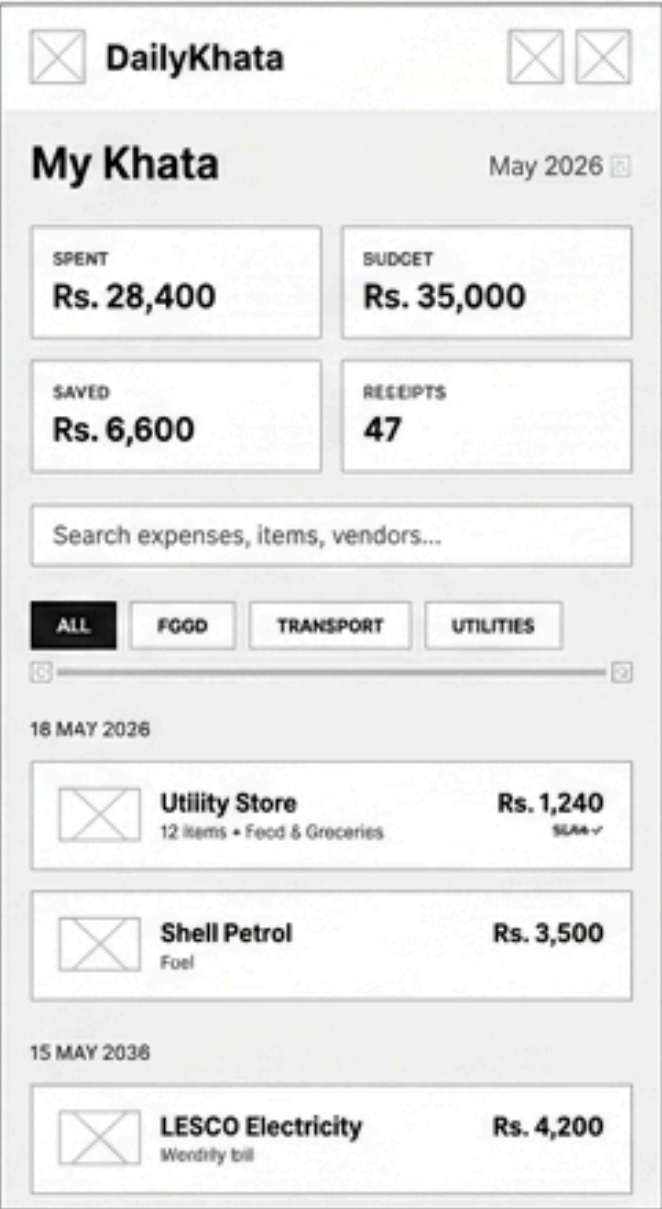
TOUCH-TO-EDIT

Tapping any flagged field allows instant correction via a numpad, providing rapid error recovery.

CONTEXTUAL ANCHORS

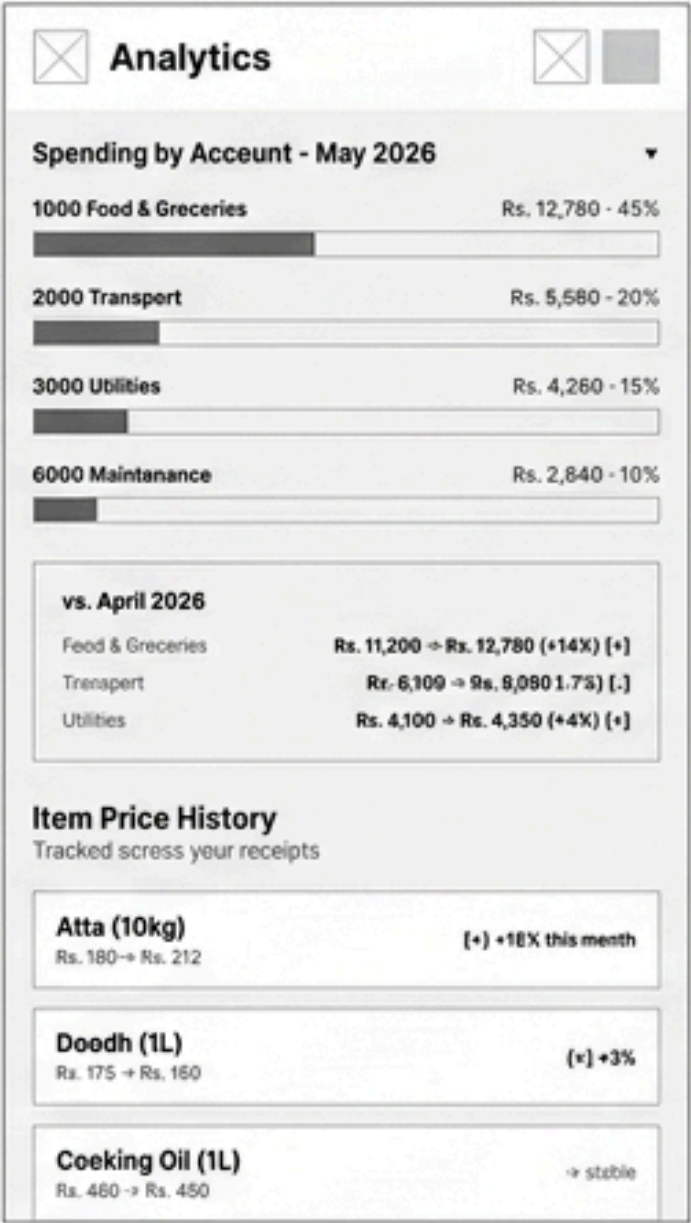
Month-over-month price changes (e.g., Atta +18%) are displayed directly on the verification screen to contextualize spending immediately.

UI ANATOMY: LEDGER, ANALYTICS, AND CONVERSATIONAL QUERY



DIGITAL KHATA [WF-04]

A full digital ledger featuring horizontal filter pills (Food, Transport, Utilities) and explicit scan verification badges per entry for traceability.



ANALYTICS [WF-05]

Avoids complex charts in favor of plain-language month-over-month changes and a dedicated item price tracker (e.g., Atta price history).



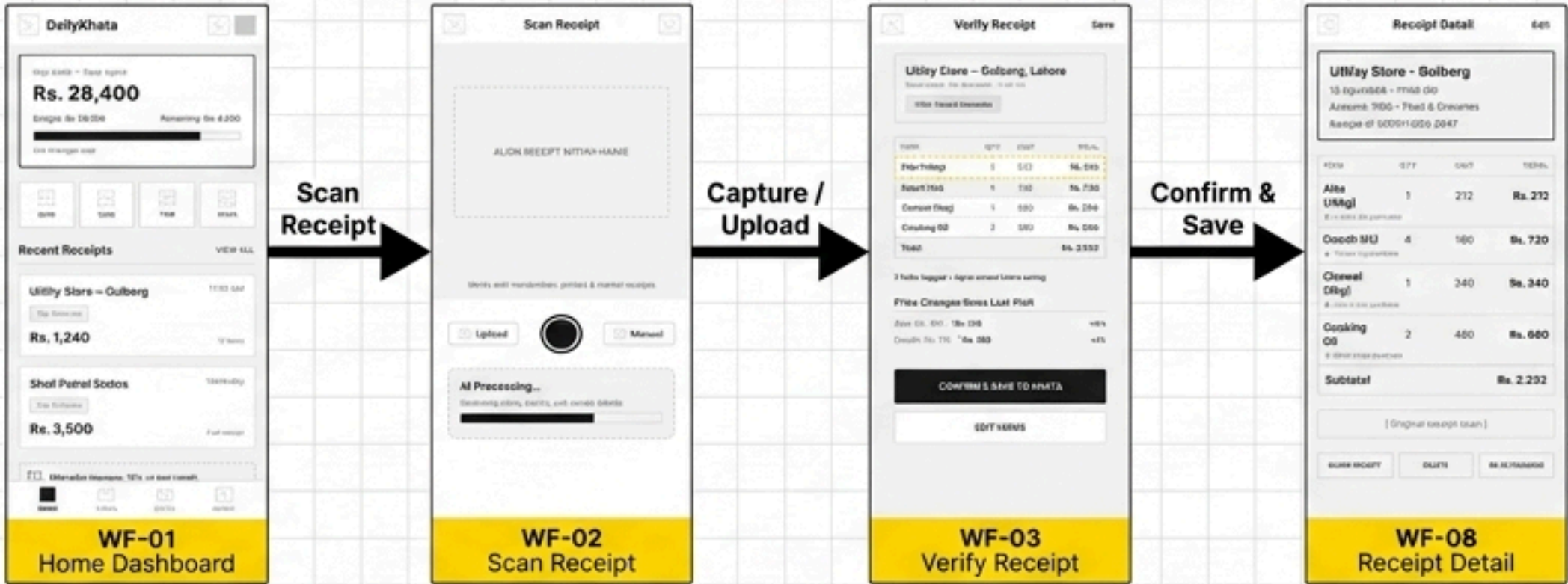
AI ASSISTANT [WF-06]

A conversational interface for complex queries. Includes predefined quick-question pills (e.g., 'Biggest expense') to support users with lower digital literacy.

CORE NAVIGATION: RECEIPT SCANNING & MANUAL ENTRY

FLOW A: SCAN & VERIFY (Task T1)

Home Dashboard -> Capture/Upload
-> Verify Flags -> Confirm & Save to
Detail View.



FLOW C: MANUAL ENTRY (Task T3)

Designed to accommodate informal,
receipt-less cash transactions
(e.g., paying a plumber).

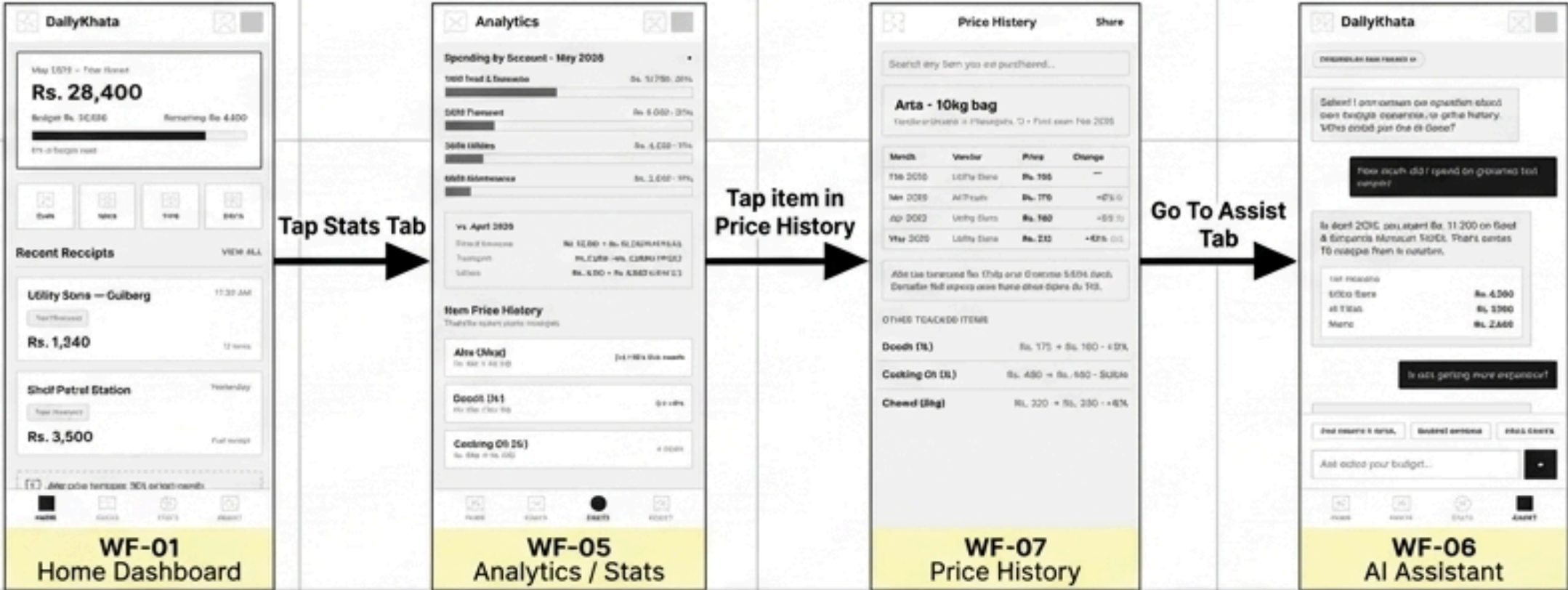
Home -> Manual Entry Form ->
Save to Ledger.



CORE NAVIGATION: HISTORICAL DATA & PROACTIVE ALERTS

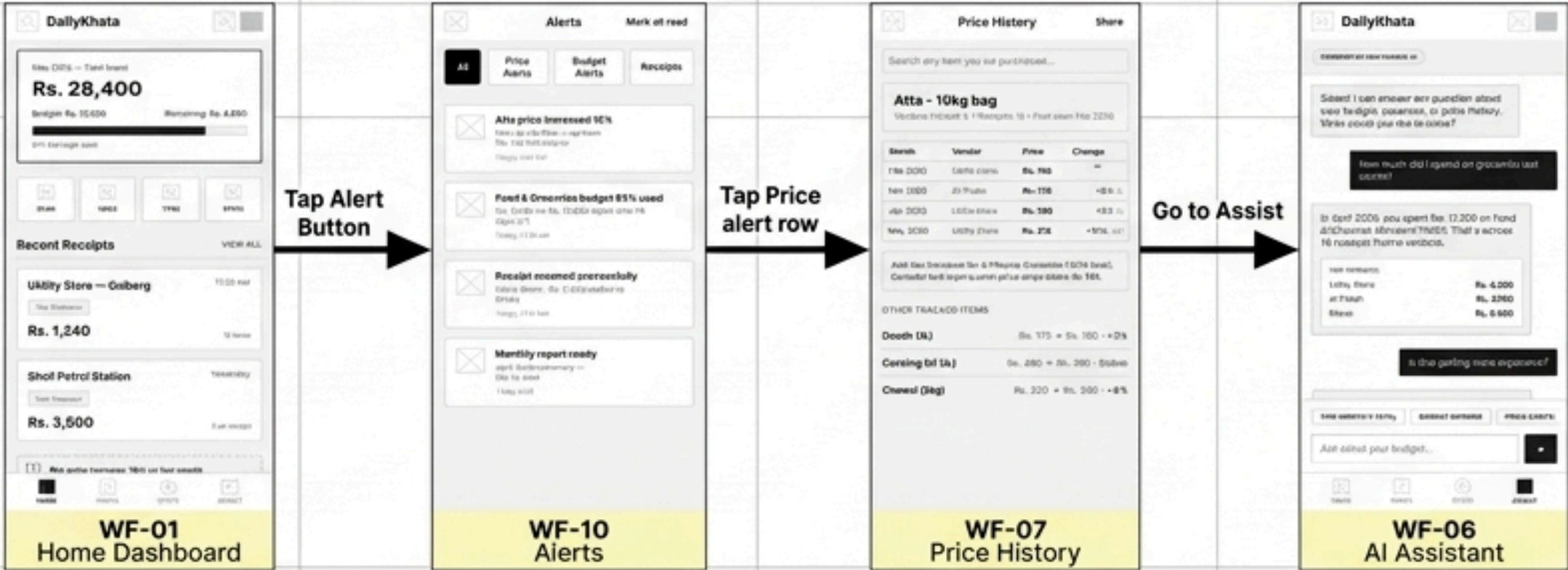
FLOW B: PRICE HISTORY (Task T2)

Analytics Tab -> Select Tracked Item
-> View Timeline -> Escalate to AI
for contextual questioning.



FLOW E: RESPOND TO ALERT (Task T6)

Notification Center -> Tap Price Alert
-> View Item History -> Ask AI for
budget adjustment recommendations.



SECTION 02

USER TEST PLAN

Empirical usability validation utilizing think-aloud protocols and the System Usability Scale (SUS).

FORMATIVE USABILITY STRATEGY

PRIMARY GOAL: Evaluate prototype navigability, clarity of AI uncertainty flags, and discoverability of conversational analytics.
TARGET SAMPLE: N=5 to 6 participants across two specific archetypes.

GROUP A: HOMEMAKERS	GROUP B: STUDENTS & FREELANCERS
Ages 28-50	Ages 18-28
Low digital literacy.	High digital literacy.
Currently relies on analogue notebooks or mental tracking.	Has tried and abandoned digital finance tools previously.
Evaluated to test the baseline accessibility and friction of the app.	Evaluated for comprehension of power-user analytics features.

THE 6 KEY USABILITY TASKS:

PERSONA-TO-TASK MATRIX

TASK	PERSONA	TARGET FLOW	OBSERVATION GOAL
T1 (Scan/Verify)	Rukhsana	Flow A	Does user understand the yellow dashed fields?
T2 (Price History)	Rukhsana	Flow B	Can they find the timeline table and read the insight card?
T3 (Manual Cash Entry)	Rukhsana	Flow C	Is the 'Maintenance' category intuitive for a plumber fee?
T4 (Ask AI)	Zaid	Flow B Variant	Do they use natural language or look for a rigid menu?
T5 (Analytics Review)	Zaid	Flow D	Is the MoM comparison card clear?
T6 (Price Alert)	Zaid	Flow E	Can they navigate from alert to actionable insight?

SESSION PROTOCOL & MODERATOR SCRIPT

00:00		00:05		00:40		00:45
00:00 WELCOME & CONSENT		THINK-ALOUD TRAINING		00:40 SUS ADMINISTRATION		00:45 DEBRIEF
Establish voluntary participation, anonymity, and recording permission.		Practice exercise using a basic calculator app to ensure participants actively narrate their internal cognitive processes.		Quantitative evaluation phase.		Open-ended discussion on expectations vs reality.
			00:10 TASK EXECUTION			
			Read standardized prompts. Strict moderator rules: No hints unless stuck for >90s (use neutral prompt: 'What would you try next?'). Max 5 mins per task.			

EMPIRICAL LOGGING & DATA CAPTURE

LOGGING SHEET

Task	Time	Success (Y/N/P)	Critical Incidents
T1			
T2			
T3			
T4			
T5			
T6			

QUANTITATIVE METRICS

Task Time (mm:ss) measured explicitly from prompt completion to task resolution or abandonment.

SUCCESS STATES

Classified strictly as Y (Success), N (Failure), P (Partial Success - meaning the user required a neutral prompt or experienced significant difficulty).

QUALITATIVE CAPTURE (CRITICAL INCIDENTS)

Logged sequentially. Defined as any observable event that meaningfully helps or hinders task completion.

Severity Scale: 1 = Minor confusion, 3 = Significant delay, 5 = Complete breakdown/task failure.

POST-TEST EVALUATION & ETHICAL STANDARDS

SYSTEM USABILITY SCALE (SUS)	
A validated 10-item Likert questionnaire (1-5 scale) administered immediately post-tasks.	12345 Strongly Disagree Strongly Agree
Provides a single benchmark score (0-100) to quantify baseline satisfaction and complexity.	

INFORMED CONSENT FRAMEWORK		
TRANSPARENCY: Purpose of study clearly stated to all participants.	PRIVACY: All data strictly anonymized (e.g., users referred to as P01, P02).	AGENCY: Participation is strictly voluntary; users retain the right to skip tasks or withdraw at any time without penalty.

THANK YOU
